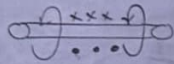


بسم الله الرحمن الرحيم

Q7



$$L = 0.5 \text{ m}$$

$$mg = 1 \text{ N} \rightarrow F_B$$

$$B = 1.5 \text{ T (outward)}$$

Find the current, $I = ?$

$$F_B = ILB$$

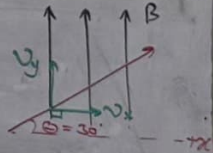
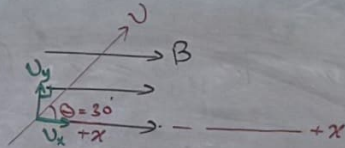
$$I = \frac{F_B}{LB} = \frac{1}{0.5 \times 1.5} = 1.33 \text{ A (left)}$$

⑧ Proton

$$B = 0.42 \text{ T (right)}$$

$$v = 2 \times 10^6 \text{ m/s}$$

$$\theta = 30^\circ \text{ at } +x$$



The distance travelled in one cycle is $2\pi r$ (circumference) 9.11×10^{-27} kg

$$T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.42} = 1.56 \times 10^{-7} \text{ s}$$

$$\begin{aligned} x &= v \cos \theta \times T \\ &= 2 \times 10^6 \cos 30^\circ \times 1.56 \times 10^{-7} = 0.27 \text{ m} \end{aligned}$$

مسألة الفيزياء

10

Proton:

$$KE = 10 \text{ MeV} = 10 \times 10^6 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-13} \text{ J}$$

$$r = 5.8 \times 10^{-10} \text{ m}$$

B = ?

$$v = \sqrt{\frac{2 KE}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-13}}{1.67 \times 10^{-27}}} = 4.38 \times 10^7$$

$$\frac{q v}{r} = m v$$

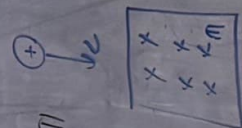
$$B = \frac{m v}{\frac{q}{r}} = \frac{1.67 \times 10^{-27} \times 4.38 \times 10^7}{1.6 \times 10^{-19} \times 5.8 \times 10^{-10}} = 7.88 \times 10^{-12} \text{ T}$$



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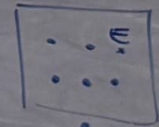
An electron is accelerated from rest through 120V. The speed is:

$$v = \sqrt{\frac{2 q V_p}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 120}{9.11 \times 10^{-31}}} = 6.5 \times 10^6 \text{ m/s}$$



$\ominus$: inward

$+$: F: inward



$\ominus$: outward

$+$: F: inward